IS Audit & Control

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Audit Planning and Analytical Procedures

Objectives

- Discuss why adequate audit planning is essential.
- Make client acceptance decisions and perform initial audit planning.
- Gain an understanding of the client's business and industry.
- Assess client business risk.
- Perform preliminary analytical procedures.
- State the purposes of analytical procedures and the timing of each purpose.
- Select the most appropriate analytical procedure from
- among the five major types.
- Compute common financial ratios.

Learning Objective 1

Discuss why adequate audit planning is essential.

Three Main Reasons for Planning

1. To obtain sufficient appropriate evidence for the circumstances
2. To help keep audit costs reasonable
3. To avoid misunderstanding with the client

8 parts for Audit Planning

FIGURE 8-1 Planning an Audit and Designing an Audit Approach



Risk Terms

- **Acceptable audit risk**
- measure of how willing the auditor is to accept that the financial statements may be materially misstated after the audit is completed.

- **Inherent risk**
- measure of likelihood that there are material misstatements in an account balance before considering the effectiveness of internal control.

Learning Objective 2

Make client acceptance decisions and perform initial audit planning.

Initial Audit Planning

Initial audit planning involves four things:

1. Client acceptance or continuance. – experienced auditor.
2. Identify client's reasons for audit.
3. Obtain an understanding with the client. (terms of the engagement)
4. Develop overall audit strategy. (including engagement staffing and any required audit specialists.)

Client Acceptance and Continuance

- New client investigations
  - If previously audited by CPA firm, new auditor is required to communicate with the predecessor auditor
  - Client permission required (Code of Professional Conduct)
- Continuing clients
  - Annual evaluations whether to continue based on issues, fees, and client integrity

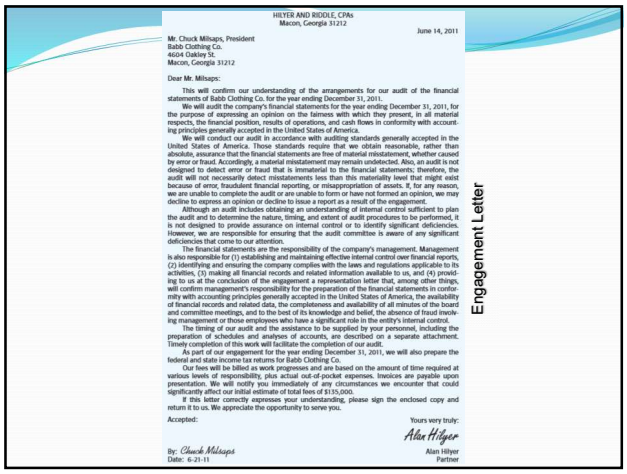
Identify Reasons for the Audit

- Two major factors affecting acceptable risk
  - Likely statement users
  - Intended uses of the statements

- Likely to accumulate more evidence for companies that are
  - Publicly held
  - Have extreme indebtedness
  - Likely to be sold

Obtaining an Understanding with the Client

- Engagement terms should be understood between CPA and client.
- Standards require an engagement letter describing:
  - objectives
  - responsibilities of auditor and management
  - schedules and fees
- Informs client that auditor cannot guarantee all acts of fraud will be discovered
- See figure (Engagement Letter)



HILYER AND RIDDLE, CPAs
Macon, Georgia 31212

June 14, 2011

Mr. Chuck Milsaps, President
Babb Clothing Co.
4604 Oakley St.
Macon, Georgia 31212

Dear Mr. Milsaps:

This will confirm our understanding of the arrangements for our audit of the financial statements of Babb Clothing Co. for the year ending December 31, 2011.

We will audit the company's financial statements for the year ending December 31, 2011, for the purpose of expressing an opinion on the fairness with which they present, in all material respects, the financial position, results of operations, and cash flows in conformity with accounting principles generally accepted in the United States of America.

We will conduct our audit in accordance with auditing standards generally accepted in the United States of America. Those standards require that we obtain reasonable, rather than absolute, assurance that the financial statements are free of material misstatement, whether caused by error or fraud. Accordingly, a material misstatement may remain undetected. Also, an audit is not designed to detect error or fraud that is immaterial to the financial statements; therefore, the audit will not necessarily detect misstatements less than this materiality level that might exist because of error, fraudulent financial reporting, or misappropriation of assets. If, for any reason, we are unable to complete the audit or are unable to form or have not formed an opinion, we may decline to express an opinion or decline to issue a report as a result of the engagement.

Although an audit includes obtaining an understanding of internal control sufficient to plan the audit and to determine the nature, timing, and extent of audit procedures to be performed, it is not designed to provide assurance on internal control or to identify significant deficiencies. However, we are responsible for ensuring that the audit committee is aware of any significant deficiencies that come to our attention.

The financial statements are the responsibility of the company's management. Management is also responsible for (1) establishing and maintaining effective internal control over financial reports, (2) identifying and ensuring the company complies with the laws and regulations applicable to its activities, (3) making all financial records and related information available to us, and (4) providing to us at the conclusion of the engagement a representation letter that, among other things, will confirm management's responsibility for the preparation of the financial statements in conformity with accounting principles generally accepted in the United States of America, the availability of financial records and related data, the completeness and availability of all minutes of the board and committee meetings, and to the best of its knowledge and belief, the absence of fraud involving management or those employees who have a significant role in the entity's internal control.

The timing of our audit and the assistance to be supplied by your personnel, including the preparation of schedules and analyses of accounts, are described on a separate attachment. Timely completion of this work will facilitate the completion of our audit.

As part of our engagement for the year ending December 31, 2011, we will also prepare the federal and state income tax returns for Babb Clothing Co.

Our fees will be billed as work progresses and are based on the amount of time required at various levels of responsibility, plus actual out-of-pocket expenses. Invoices are payable upon presentation. We will notify you immediately of any circumstances we encounter that could significantly affect our initial estimate of total fees of $135,000.

If this letter correctly expresses your understanding, please sign the enclosed copy and return it to us. We appreciate the opportunity to serve you.

Accepted:

By: Chuck Milsaps
Date: 6-21-11

Yours very truly:

Alan Hilyer

Alan Hilyer
Partner

Engagement Letter

Develop Overall Audit Strategy

- Preliminary audit strategy should consider
  - client's business and industry
  - material misstatement risk areas
  - number of client locations
  - past effectiveness of controls
- Preliminary strategy helps auditor determine resource requirements and staffing
  - staff continuity
  - need for specialists

Learning Objective 3

Gain an understanding of the client's business and industry.

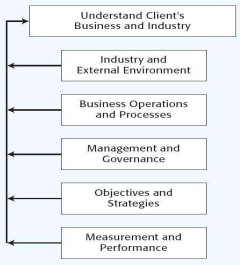
Understanding of the Client's Business and Industry

Client business risk is the risk that the client will fail to meet its objectives.

- Economic conditions around the world
- Information technology
- Clients expanded operations globally
- Human capital & intangible assets has increased accounting complexity

Understanding of the Client's Business and Industry

FIGURE 8-3 Strategic Systems Understanding of the Client's Business and Industry



Industry and External Environment

Reasons for obtaining an understanding of the client's industry and external environment:

1. Risks associated with specific industries
2. Inherent risks common to all clients in certain industries
3. Unique accounting requirements

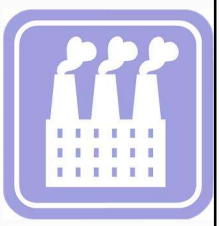
Business Operations and Processes

Factors the auditor should understand:

- Major sources of revenue
- Key customers and suppliers
- Sources of financing
- Information about related parties

Tour the Plant and Offices

Touring the physical facilities enables the auditor to assess asset safeguards and interpret accounting data related to assets.



Identify Related Parties

- Affiliated companies
- Principal owners of the client
- Any other party with which the client deals
- A party who can influence management or client policies

Management and Governance

Governance includes:
- Organizational structure
- Board activities
- Audit committee activities.

Governance insights:
- Corporate charter and bylaws
- Code of ethics
- Meeting minutes

Management establishes the strategies and processes followed by the client's business.

Code of Ethics

In response to the Sarbanes-Oxley Act, the SEC now requires each public company to disclose whether is has adopted a code of ethics that applies to senior management.

The SEC also requires companies to disclose amendments and waivers to the code of ethics.

Client Objectives and Strategies

Strategies are approaches to achieve organizational objectives.

Auditors should understand client objectives.

- Financial reporting reliability
- Effectiveness and efficiency of operations
- Compliance with laws and regulations

Measurement and Performance

The client's performance measurement system includes key performance indicators. Examples:

- market share
- sales per employee
- unit sales growth
- Web site visitors
- same-store sales
- sales/square foot

Performance measurement includes ratio analysis and benchmarking against key competitors.

Learning Objective 4

Assess client business risk.

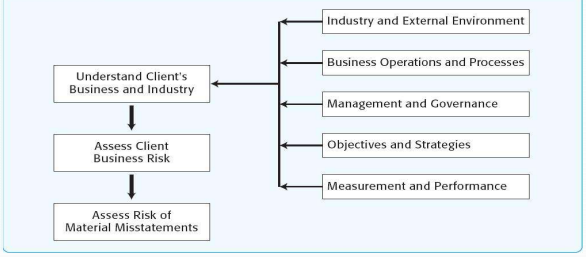
Assess Client Business Risk

Client business risk is the risk that the client will fail to achieve its objectives.

- What is the auditor's primary concern?
- Material misstatements in the financial statements due to client business risk

Client's Business, Risk, and Risk of Material Misstatement

FIGURE 8-4 **Understanding the Client's Business and Industry, Client Business Risk, and Risk of Material Misstatement**



Sarbanes-Oxley Act

Management must certify it has designed disclosure controls and procedures to ensure that material information about business risks is made known to them.

Management must certify it has informed the auditor and audit committee of any significant control deficiencies.

Learning Objective 5

Perform preliminary analytical procedures.

Preliminary Analytical Procedures

Comparison of client ratios to industry or competitor benchmarks provides an indication of the company's performance.

Preliminary tests can reveal unusual changes in ratios.

Examples of Planning Analytical Procedures

TABLE 8-1 Examples of Planning Analytical Procedures

Selected Ratios	Hillsburg 12/31/11	Industry 12/31/11	Hillsburg 12/31/10	Industry 12/31/10
Short-Term Debt-Paying Ability				
Cash ratio	0.06	0.22	0.06	0.20
Quick ratio	1.57	3.10	1.45	3.00
Current ratio	3.86	5.20	4.04	5.10
Liquidity Activity Ratios				
Accounts receivable turnover	7.59	12.15	7.61	12.25
Days to collect accounts receivable	48.09	30.04	47.96	29.80
Inventory turnover	3.36	5.20	3.02	4.90
Days to sell inventory	108.63	70.19	120.86	74.49
Ability to Meet Long-Term Obligations				
Debt to equity	1.73	2.51	1.98	2.53
Times interest earned	3.06	5.50	3.29	5.60
Profitability Ratios				
Gross profit percent	27.85	31.00	27.70	32.00
Profit margin	0.05	0.07	0.05	0.08
Return on assets	0.09	0.09	0.08	0.09
Return on common equity	0.26	0.37	0.24	0.35

Summary of the Parts of Auditing Planning

A major purpose is to gain an understanding of the client's business and industry.

Planning an Audit and Designing an Audit Approach

- Set materiality and assess acceptable audit risk and inherent risk.
- Understand internal control and assess control risk



- Gather information to assess fraud risks
- Develop overall audit plan and audit program

Learning Objective 6

State the purposes of analytical procedures and the timing of each procedure.

Analytical Procedures

AU 329 emphasizes the expectations developed by the auditor.

1. Required in the planning phase
2. Often done during the testing phase
3. Required during the completion phase

Timing and Purposes of Analytical Procedures

FIGURE 8-6 Timing and Purposes of Analytical Procedures

Purpose	Phase		
	(Required) Planning Phase	Testing Phase	(Required) Completion Phase
Understand the client's business and industry	Primary purpose		
Assess going concern	Secondary purpose		Secondary purpose
Indicate possible misstatements (attention directing)	Primary purpose	Secondary purpose	Primary purpose
Reduce detailed tests	Secondary purpose	Primary purpose	

Learning Objective 7

Select the most appropriate analytical procedure from among the five major types.

Five Types of Analytical Procedures

Compare client data with:

1. Industry data
2. Similar prior-period data
3. Client-determined expected results
4. Auditor-determined expected results
5. Expected results using nonfinancial data.

Compare Client and Industry Data

	Client		Industry	
	2009	2008	2009	2008
Inventory turnover	3.4	3.5	3.9	3.4
Gross margin	26.3%	26.4%	27.3%	26.2%

Internal Comparisons

TABLE 8-2 Internal Comparisons and Relationships

Ratio or Comparison	Possible Misstatement
Raw material turnover for a manufacturing company	Misstatement of inventory or cost of goods sold or obsolescence of raw material inventory
Sales commissions divided by net sales	Misstatement of sales commissions
Sales returns and allowances divided by gross sales	Misclassified sales returns and allowances or unrecorded returns or allowances subsequent to year-end
Bad debt expense divided by net sales	Misstatement in the allowance for bad debts
Each of the individual manufacturing expenses as a percent of total manufacturing expense	Significant misstatement of individual expenses within a total

Compare Client Data with Similar Prior Period Data

	2009		2008	
	(000) Prelim.	% of Net sales	(000) Prelim.	% of Net sales
Net sales	$143,086	100.0	$131,226	100.0
Cost of goods sold	103,241	72.1	94,876	72.3
Gross profit	$ 39,845	27.9	$ 36,350	27.7
Selling expense	14,810	10.3	12,899	9.8
Administrative expense	17,665	12.4	16,757	12.8
Other	1,689	1.2	2,035	1.6
Earnings before taxes	$ 5,681	4.0	$ 4,659	3.5
Income taxes	1,747	1.2	1,465	1.1
Net income	$ 3,934	2.8	$ 3,194	2.4

Learning Objective 8

Compute common financial ratios.

Common Financial Ratios

- Short-term debt-paying ability
- Liquidity activity ratios
- Ability to meet long-term debt obligations
- Profitability ratios

Short-term Debt-paying Ability

Cash ratio = $\frac{\text{(Cash + Marketable securities)}}{\text{Current liabilities}}$

Quick ratio = $\frac{\text{(Cash + Marketable securities + Net accounts receivable)}}{\text{Current liabilities}}$

Current ratio = $\frac{\text{Current assets}}{\text{Current liabilities}}$

Liquidity Activity Ratios

Accounts receivable turnover = $\frac{\text{Net sales}}{\text{Average gross receivables}}$

Days to collect receivable = $\frac{\text{365 days}}{\text{Accounts receivable turnover}}$

Inventory turnover = $\frac{\text{Cost of goods sold}}{\text{Average inventory}}$

Days to sell inventory = $\frac{\text{365 days}}{\text{Inventory turnover}}$

Ability to Meet Long-term Debt Obligation

Debt to equity = $\frac{\text{Total liabilities}}{\text{Total equity}}$

Times interest earned = $\frac{\text{Operating income}}{\text{Interest expense}}$

Profitability Ratios

Earnings per share = $\frac{\text{Net income}}{\text{Average common shares outstanding}}$

Gross profit percent = $\frac{\text{(Net sales – Cost of goods sold)}}{\text{Net sales}}$

Profit margin = $\frac{\text{Operating income}}{\text{Net sales}}$

Profitability Ratios

Return on assets = Income before taxes / Average total assets

Return on common equity = (Income before taxes – Preferred dividends) / Average stockholders' equity

Common financial ratios.

Explained

Common financial ratios.

- Auditors' analytical procedures includes the use of general financial ratios during planning and final review of the audited financial statements.
- These are useful for understanding recent events and the financial status of the business and for viewing the statements from the perspective of a user.
- The general financial analysis may be effective for identifying possible problem areas.
- The most important comparisons are to those of previous years for the company and to industry averages or similar companies for the same year.

Accounting terms

- **Liquidity:** Cash is the most liquid asset, while real estate, fine art and collectibles are all relatively illiquid.
- **Market liquidity:** In business, economics or investment, market liquidity is a market's ability to facilitate the purchase or sale of an asset without causing drastic change in the asset's price.
- **Leverage ratio**: A leverage ratio is any one of several financial measurements that look at how much capital comes in the form of debt (loans), or assesses the ability of a company to meet financial obligations.
- **Equity**: the value of the shares issued by a company.
  - "he owns 62% of the group's equity."

Accounting terms

- **Cash ratio:** ratio of a company's total cash + cash equivalents to its current liabilities. It is most commonly used as a measure of company liquidity.
  cash ratio = (cash + cash equivalents)/ (total current liabilities)
  - **Example:** Ally's Palace is a restaurant that is looking to remodel its dining room. Ally is asking her bank for a loan of $100,000. Ally's balance sheet lists these items:
    Cash: $10,000, Cash Equivalents: $2,000
    Accounts Payable: $5,000, Current Taxes Payable: $1,000
    Current Long-term Liabilities: $10,000
    Ally's cash ratio is calculated like this:
    CR = (10,000+2,000)/ (5,000+1,000+10,000) = 0.75
  - This means that Ally only has enough cash and equivalents to pay off 75 % of her current liabilities.

Accounting terms

- **Quick ratio:** compares the total amount of cash + marketable securities + accounts receivable to the amount of current liabilities.
  - The quick ratio is also known as the acid test ratio.
  - The quick ratio is an indicator of a company's short-term liquidity. The quick ratio measures a company's ability to meet its short-term obligations with its most liquid assets. For this reason, the ratio excludes inventories from current assets, and is calculated as follows:
  - Quick ratio = (current assets – inventories) / current liabilities, or
  - = (cash and equivalents + marketable securities + accounts receivable) / current liabilities

Accounting terms

- Current ratio: The current ratio is a liquidity ratio that measures a company's ability to pay short-term and long-term obligations. To gauge this ability, the current ratio considers the total assets of a company (both liquid and illiquid) relative to that company's total liabilities.
  - Current Ratio = Current Assets / Current Liabilities
  - The current ratio is called "current" because, unlike some other liquidity ratios, it incorporates all current assets and liabilities.

Accounting terms

- **Accounts receivable** (AR): refers to money owed by customers (individuals or corporations) to another entity in exchange for goods or services that have been delivered or used, but not yet paid for.
- **Accounts payable** (AP): is an accounting entry that represents an entity's obligation to pay off a short-term debt to its creditors. Accounts payable entry is found on balance sheet under the heading current liabilities.
- **Debt - Equity Ratio**: indicates how much debt a company is using to finance its assets relative to the amount of value represented in shareholders' equity.
  - D-E Ratio = Total Liabilities / Shareholders' Equity

Accounting terms

- **Times interest earned: (TIE)** a metric used to measure company's ability to meet its debt obligations.
  TIE = (earnings before interest and taxes (EBIT))/ (total interest payable on bonds & contractual debt).
- TIE = Operating income / Interest expense.
- It indicates how many times a company can cover its interest charges on a pretax basis. Failing to meet these obligations could force a company into bankruptcy.
- **Gross profit:** Gross profit is a company's total revenue (equivalent to total sales) minus the cost of goods sold. Gross profit is the profit a company makes after deducting the costs associated with making and selling its products, or the costs associated with providing its services. formula: Gross profit = revenue (total sales) - cost of goods sold

Accounting terms

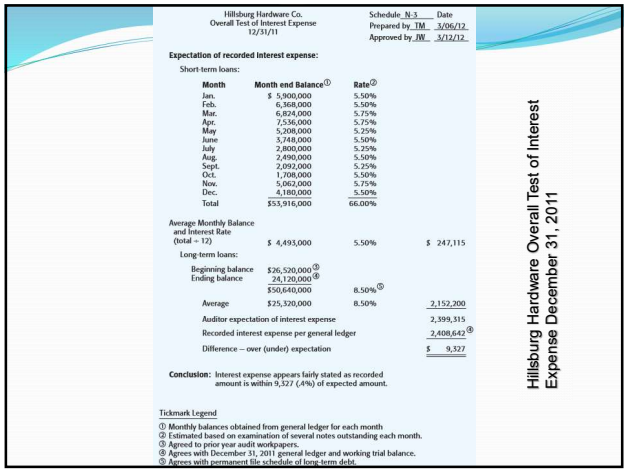
- **Net profit:** No. of sales dollars remaining after all operating expenses, interest, taxes and preferred stock dividends have been deducted from total revenue.
- (Example): Net profit is also referred to as the bottom line, net income, or net earnings. The formula for net profit is as follows:
  Total Revenue -Total Expenses = Net Profit
- Net profit is found on the last line of the income statement, which is why it's often referred to as the bottom line. Let's look at a hypothetical income statement for Company XYZ:

Income Statement of XYZ, Inc. - December 31, 2008:	
Total Revenue (Net income)	$100,000
Cost of Goods Sold	$ 20,000
Gross Profit	$ 80,000
Operating Expenses	
Salaries	$10,000
Rent	$10,000
Utilities	$5,000
Depreciation	$5,000
Total Operating Expenses	$30,000
Interest Expense	$ 10,000
Taxes	$ 10,000

Net Profit = $100,000 - $20,000 - $30,000, - $10,000 - $10,000 = $30,000

Accounting terms

- **Profit Margin** = Net Income / Net Sales (revenue)
- **Return on equity (ROE):** is the amount of net income returned as a percentage of shareholders equity. Return on equity measures a corporation's profitability by revealing how much profit a company generates with the money shareholders have invested.
  - Return on Equity = Net Income/Shareholder's Equity
- **Return on common equity (ROCE)**: can be defined as the amount of profit or net income a company earns per investment dollar. Return on common equity, explained is a measure of how well a company uses its investment dollars to generate profits.
  - ROCE = Net Income (NI)/ Average Common Shareholder's Equity
  - The average common equity is found by combining the beginning common stock for the year on the balance sheet, and the ending common stock value. These values are then divided by two for the average amount in the year.



Hillsburg Hardware Co.
Overall Test of Interest Expense
12/31/11

Schedule N-3 Date
Prepared by TM 3/06/12
Approved by JW 3/12/12

Expectation of recorded interest expense:

Short-term loans:

Month	Month end Balance①	Rate②	
Jan.	$ 5,900,000	5.50%	
Feb.	6,368,000	5.50%	
Mar.	6,824,000	5.75%	
Apr.	7,536,000	5.75%	
May	5,208,000	5.25%	
June	5,748,000	5.50%	
July	2,800,000	5.25%	
Aug.	2,490,000	5.50%	
Sept.	2,092,000	5.25%	
Oct.	1,708,000	5.50%	
Nov.	5,062,000	5.75%	
Dec.	4,180,000	5.50%	
Total	$53,916,000	66.00%	
Average Monthly Balance and Interest Rate (total ÷ 12)	$ 4,493,000	5.50%	$ 247,115

Long-term loans:

Beginning balance	$26,520,000③		
Ending balance	24,120,000④		
	$50,640,000	8.50%⑤	
Average	$25,320,000	8.50%	2,152,200
Auditor expectation of interest expense			2,399,315
Recorded interest expense per general ledger			2,408,642⑥
Difference – over (under) expectation			$ 9,327

Conclusion: Interest expense appears fairly stated as recorded amount is within 9,327 (.4%) of expected amount.

Tickmark Legend

① Monthly balances obtained from general ledger for each month
② Estimated based on examination of several notes outstanding each month.
③ Agreed to prior year audit workpapers.
④ Agrees with December 31, 2011 general ledger and working trial balance.
⑤ Agrees with permanent file schedule of long-term debt.

Hillsburg Hardware Overall Test of Interest Expense December 31, 2011

- Short-term Debt-Paying Ability

Cash ratio =	$\frac{\text{cash + marketable securities}}{\text{current liabilities}}$	$\frac{828}{13,216} = 0.06$
Quick ratio =	$\frac{\text{cash + marketable securities + net accounts receivable}}{\text{current liabilities}}$	$\frac{828 + 18,957 + 945}{13,216} = 1.57$
Current ratio =	$\frac{\text{current assets}}{\text{current liabilities}}$	$\frac{51,027}{13,216} = 3.86$

- Liquidity Activity Ratios

Accounts receivable turnover =	$\frac{\text{net sales}}{\text{average gross receivables}}$	$\frac{143,086}{((18,957 + 1,240) + (16,210 + 1,311))/2} = 7.59$
Days to collect receivables =	$\frac{365 \text{ days}}{\text{accounts receivable turnover}}$	$\frac{365 \text{ days}}{7.59} = 48.09 \text{ days}$
Inventory turnover =	$\frac{\text{cost of goods sold}}{\text{average inventory}}$	$\frac{103,241}{(29,865 + 31,600)/2} = 3.36$
Days to sell inventory =	$\frac{365 \text{ days}}{\text{inventory turnover}}$	$\frac{365 \text{ days}}{3.36} = 108.63 \text{ days}$

Summary of Analytical Procedures

Compare ratios of recorded amounts to auditor expectations.

Used in planning to understand client's business and industry.

Used throughout the audit
- to identify possible misstatements
- reduce detailed tests
- assess going-concern issues.